

Write legibly showing all the steps VERY clearly. Max. score=100 (Total points = 105).

- (1) (a) Define the following: • semi-algebra, • algebra, • σ -algebra.
 (b) Define the Δ operation (symmetric difference) between two sets E and F as:
 $E\Delta F = (E \cap F^c) \cup (E^c \cap F)$ set of elements that are in either E or F , but not in both.
 Let $\mathcal{F}$ be a sigma algebra of subsets of Ω and μ be a measure on that.
 Then show: for all $A, B, C \in \mathcal{F}$, $\mu(A\Delta C) \leq \mu(A\Delta B) + \mu(B\Delta C)$.
 (5×3 + 10 = 25 points)
- (2) (a) Let $\mathcal{F}$ be a σ -algebra of subsets of $\Omega = \mathbb{R}$. Take $\Omega_0 = [0, \infty)$ and define $\mathcal{F}_0 = \{F \cap [0, \infty) : F \in \mathcal{F}\}$. Show that $\mathcal{F}_0$ is a σ -algebra of subsets of $\Omega_0 = [0, \infty)$.
 (b) Let $\Omega \neq \emptyset$ and $\mathcal{P}(\Omega)$ be the power set. Let μ^* be an *outer*-measure defined on *any* $A \in \mathcal{P}(\Omega)$ and $\mathcal{M}$ be the collection of μ^* -measurable sets.
 Show that if $\mu^*(A) = 0$, then $A \in \mathcal{M}$.
 (Hint: Need 3 defining properties of outer-measure & definition of μ^* -measurable sets).
 (10 × 2 = 20 points)
- (3) Let $\Omega = \mathbb{N}$, $\mathcal{F} = \mathcal{P}(\mathbb{N})$. Define $A_n = \{j : j \geq n\}$, $n \geq 1$.
 (a) If $\mu(A) = |A|$ (counting measure). Show that monotone continuity from above does **not** hold for this sequence of decreasing $\{A_n\} \subset \mathcal{F}$.
 (b) On the same $(\Omega, \mathcal{F})$, can you specify a *different* measure μ' such that the monotone continuity from above does hold for the *same* $\{A_n\} \subset \mathcal{F}$?
 (c) In general, under what conditions (on A_n 's and μ) does the monotone continuity from *above* hold?
 (d) In general, under what conditions (on A_n 's and μ) does the monotone continuity from *below* hold?
 (4 × 10 = 40 points)
- (4) Take $F(x) = (1 - e^{-x}) \mathbf{I}_{\{x \geq 0\}}$.
 (a) Using the steps in Caratheodory Extension Theorem(s), show by starting from defining a measure μ_F on a suitable semi-algebra, how one can extend it to the Lebesgue-Stieltjes measure μ_F^* on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.
 [Write down the steps of the theorems carefully and state the results used in each step]
 (b) Is such a measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ unique? Why?
 (c) If $A = \{x \in \mathbb{R} : |x| > 2\}$, then compute $\mu_F^*(A)$.
 (5 × 3 = 15 points)